

Shivam Kamboj

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Portfolio | GitHub | LinkedIn

EDUCATION

University of California, Merced

Ph.D. in Physics; advisor: Prof. Lin Tian

Research: Quantum Rabi systems, bosonic qubits, and superconducting-circuit modeling.

Merced, CA

Expected Fall 2026

San Jose State University

M.S. in Physics

San Jose, CA

2022

RESEARCH EXPERIENCE

Tian Group, University of California, Merced

Graduate Student Researcher

Merced, CA

May 2023 – Present

- Characterized a regime of the quantum Rabi dimer in which an isolated even–odd cat-state doublet forms a candidate bosonic-qubit codespace under a static Hamiltonian, without continuous driving or engineered dissipation.
- Combined normal-mode analysis and sparse exact diagonalization to map the low-energy spectrum, distinguish crossings by parity and site-exchange symmetry, and track eigenstates as the inter-cavity coupling changes.
- Studied state evolution, entanglement, and photon-loss dynamics to distinguish coherent motion within the doublet from decoherence and leakage. Developed Python/QuTiP workflows with convergence checks and checkpointed SLURM runs.
- Built COMSOL electromagnetic models of a proposed superconducting implementation. Extracted capacitances, mode frequencies, impedances, and inductive participation to assess circuit feasibility and identify constraints on the required coupling.

Lawrence Livermore National Laboratory

Physics Graduate Intern, Quantum Coherent Device Physics Group

Livermore, CA

May – Aug 2024

- Developed Python software for pulse-level control of superconducting qudits, designed and evaluated gate sequences, and characterized their performance using quantum process tomography.
- Built a generalized linear-inversion tomography pipeline using pseudo-inverse reconstruction to recover process maps from experimental data.
- Automated Ramsey and T_1/T_2 fitting to characterize device behavior and provide feedback for control experiments.

Sharping Group, University of California, Merced

Research Collaborator, Experimental Quantum Optics

Merced, CA

2024 – Present

- Developed Hamiltonian models of coupled YIG–cavity systems to study magnon–photon spectra and entanglement, including the effects of inter-cavity coupling and parametric driving.

Hurst Group, San Jose State University and UC Merced

Graduate Researcher

San Jose / Merced, CA

Jun 2021 – Jan 2024

- Derived analytical dispersion relations and developed numerical solvers for two-dimensional spin-torque oscillator arrays. Studied oscillatory edge modes and their robustness to disorder, leading to a first-author publication in *Physical Review B*.

PUBLICATIONS AND MANUSCRIPTS

1. **S. Kamboj**, R. A. Duine, B. Flebus, and H. M. Hurst. “Oscillatory edge modes in two-dimensional spin-torque oscillator arrays.” *Physical Review B* **109**, 094436 (2024).
doi:10.1103/PhysRevB.109.094436.
2. **S. Kamboj** and L. Tian. “Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer.” Manuscript in preparation.

SCIENTIFIC SOFTWARE AND PROJECTS

nonmarkov_qec

Open-source Python package; released on PyPI

2026 – Present

- Built a surface-code benchmarking pipeline that combines stochastic noise generation, Stim simulation, PyMatching decoding, and Monte Carlo sampling with bootstrap uncertainty estimates.
- Compared temporally correlated data-gate dephasing with a statistically matched white-noise baseline. For distances 3 and 5, the estimated crossing remained consistent within uncertainty over correlation times of 0.2–200 cycles under the tested noise model.
- Maintained 74 tests, strict type checking, continuous integration, and automated package releases, with documented methods and reproducible benchmark scripts.

Physics Concept Analyzer Tool (PCAT)

Developer; Python, Flask, Gemini API, Google Cloud

UC Merced

2025

- Designed and deployed a physics-teaching application on Google App Engine. The tool guides students through concept identification, assumptions, and model construction before calculation, with support for text, voice, and document inputs.

TEACHING EXPERIENCE

University of California, Merced and San Jose State University

Teaching Assistant, Physics

California

2021 – Present

- Led laboratory and discussion activities, held office hours, and assessed student work across physics courses. In introductory laboratories for biological science students, connected measurements and uncertainty to physical models.
- Collaborated with faculty on inquiry-based laboratory exercises and piloted PCAT to support students' reasoning during problem solving.
- Received teaching awards at UC Merced (2026) and San Jose State University (2022).

Guest lecturer, Computational Physics (PHYS 281), UC Merced

2024, 2026

SELECTED PRESENTATIONS

- APS Global Physics Summit, 2026.
- Poster presentation, PLS Division Student Poster Symposium, LLNL, 2024.
- Invited talk, "Simulating Quantum Systems," Friday Seminar Series, CSU Stanislaus, 2024.
- S. Kamboj and H. M. Hurst, "Topology of a Driven 2D Spin Torque Oscillator Array," APS March Meeting, Chicago, 2022. Abstract T00.249.

HONORS AND AWARDS

Outstanding Graduate Teaching Award, UC Merced	2026
Physics Graduate Group Summer Fellowship, UC Merced (\$7,000 each)	2025, 2026
Physics Graduate Group Travel Fellowship, UC Merced (\$750)	2026
Outstanding Teaching Assistant Award, San Jose State University (\$1,000)	2022
First place, Tech-Invent competition; silicon particle detector presentation	2017

SERVICE, LEADERSHIP, AND OUTREACH

Physics Delegate, Graduate Student Association, UC Merced	2024
Member, Graduate Dean Advisory Council on Diversity, UC Merced	2023
Mentor, Learning Assistant Reflection, Guidance and Exploration program	2022

TECHNICAL SKILLS

Quantum theory and numerics: Hamiltonian modeling, normal-mode and Bogoliubov analysis, symmetry-resolved spectra, sparse exact diagonalization, open-system dynamics, and convergence analysis.

Circuit modeling and characterization: COMSOL RF and AC/DC; capacitance, impedance, and inductive participation extraction; pulse-level control; process tomography; Ramsey and T_1/T_2 analysis.

Scientific software: Python, NumPy, SciPy, QuTiP, Matplotlib, Stim, PyMatching, Qiskit, Flask; Git, Linux, Bash, SLURM, pytest, ruff, mypy, Google Cloud, and $\text{\LaTeX}$.